Week 5

Spring 2027 · Section 029 · MW 9:00–10:20 AM · RRC Room 8210

Course Level Objectives:

- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)
- Solve linear regression and correlation problems. (CLO 7)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compare a quantitative variable across groups using side-by-side boxplots. (CLO 2) **2.** Compute and interpret a difference in means. (CLO 2) **3.** Describe the direction, form, and strength of a relationship in a scatterplot. (CLO 7) **4.** Estimate the correlation r from a scatterplot, and compute it using technology. (CLO 7) **5.** Explain how outliers and nonlinear patterns affect r. (CLO 7) **6.** Explain why correlation doesn't imply causation. (CLO 7)	e-Text: 2.4 Two Variable Relationships Slides: 2.4 Two Variable Relationships	Participation: Box Plots and Outliers – 1.5×IQR rule review; side-by-side boxplots (tea vs. coffee) Your Turn: Tea vs. Coffee: Descriptive Statistics Participation: Two Variable Relationships – NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier	Online Homework: 2.4 Two Variable Relationships (due Sun Feb 21, 11:59 PM) Take-Home Quiz 3 (due Wed Feb 24, 11:59 PM) – covers 2.4–2.5 (all objectives) Test 1 (Fri Feb 26 – Sun Feb 28, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Feb 25) (all objectives)	WileyPLUS Blackboard Respondus
7. Interpret the slope of a regression line in context with correct units. (CLO 7) **8.** Interpret the intercept of a regression line in context and evaluate whether it is meaningful. (CLO 7) **9.** Use a regression equation to predict a response value. (CLO 7) **10.** Compute and interpret a residual. (CLO 7)	e-Text: 2.5 Two Quantitative Variables: Linear Regression Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1) Slides: 2.5 Two Quantitative Variables: Linear Regression	Examples: baseball game length and commute-time regression lines (predict, interpret slope, find a residual)	Take-Home Quiz 3 (due Wed Feb 24, 11:59 PM) – covers 2.4–2.5 (all objectives) Test 1 (Fri Feb 26 – Sun Feb 28, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Feb 25) (all objectives)	WileyPLUS StatKey Blackboard Respondus